

RIYADH KING KHALED INTL, Saudi Arabia

WMO: 404370

Lat: 24.958N

Lon: 46.699E

Elev: 625

StdP: 94.04

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.8	(c) 6.2	(d) -14.0	(e) 1.2	(f) 17.4	(g) -11.8	(h) 1.5	(i) 19.5	(j) 9.1	(k) 16.8	(l) 8.3	(m) 17.3	(n) 1.5	(o) 300	(p) 0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	44.9	18.9	44.0	18.6	43.1	18.4	21.0	38.3	20.2	37.7	19.5	37.9	4.3	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	13.3	22.1	16.1	12.3	21.4	15.0	11.5	21.4	63.8	38.0	60.6	37.5	58.1	37.4	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.9	(b) 7.9	(c) 6.8	(e) 1.2	(f) 46.3	(g) 2.1	(h) 0.8	(i) -0.2	(j) 46.8	(k) -1.4	(l) 47.3	(m) -2.6	(n) 47.7	(o) -4.1	(p) 48.3		
(a) 8.9	(b) 7.9	(c) 6.8	(e) -2.0	(f) 22.9	(g) 2.0	(h) 1.2	(i) -3.4	(j) 23.8	(k) -4.6	(l) 24.5	(m) -5.7	(n) 25.2	(o) -7.2	(p) 26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.2	14.5	17.0	21.1	26.4	32.0	34.6	35.8	35.6	32.5	27.4	20.5	15.7
	DBStd	8.20	3.31	3.65	3.46	3.31	2.35	1.55	1.48	1.41	1.96	2.28	3.30	3.28
	HDD10.0	7	4	1	0	0	0	0	0	0	0	0	0	2
	HDD18.3	311	126	62	13	1	0	0	0	0	0	0	16	93
	CDD10.0	5903	143	198	345	492	683	739	799	794	674	541	315	179
	CDD18.3	3165	6	26	99	243	424	489	541	536	424	282	81	12
	CDH23.3	50910	86	277	1048	3038	6817	8626	9753	9597	6964	3762	817	126
	CDH26.7	34649	11	72	413	1642	4567	6268	7282	7131	4777	2179	293	14
Wind	WSAvg	2.9	2.6	3.0	3.2	3.4	3.1	3.4	3.5	3.0	2.3	2.0	2.4	2.4
Precipitation	PrecAvg	116	19	8	20	25	8	0	0	0	0	1	18	17
	PrecMax	282	75	40	96	85	48	0	1	2	1	24	152	108
	PrecMin	25	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	67	20	11	28	19	13	0	0	0	0	5	32	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.0	31.2	35.2	39.4	43.3	45.2	46.0	45.6	43.8	39.2	33.2	28.2
		MCWB	14.1	15.2	15.5	17.0	18.1	18.8	19.3	19.7	18.9	17.6	16.0	14.5
	2%	DB	25.8	29.0	33.0	37.8	42.0	44.1	44.9	44.7	42.2	37.9	31.8	26.4
		MCWB	12.9	14.3	15.2	16.8	18.1	18.4	18.8	19.1	18.4	16.8	15.5	14.0
	5%	DB	23.8	26.8	31.0	36.0	40.8	43.1	44.0	43.9	41.2	36.6	30.1	24.7
		MCWB	12.1	13.4	14.7	16.5	17.9	18.0	18.5	18.8	18.1	16.5	15.1	13.5
	10%	DB	21.5	24.8	29.0	34.2	39.6	42.1	43.1	43.0	40.1	35.2	28.1	22.8
		MCWB	11.5	12.5	14.1	16.1	17.6	17.7	18.3	18.7	17.8	16.2	14.6	12.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.6	17.5	19.1	20.6	21.1	20.7	22.0	22.7	21.1	20.4	19.7	18.0
		MCDB	21.6	25.1	26.6	28.5	35.0	41.2	40.2	38.9	38.4	30.4	24.8	22.5
	2%	WB	15.1	16.0	17.7	19.3	20.0	19.4	20.3	21.1	19.7	19.0	18.1	16.5
		MCDB	20.5	23.7	26.2	28.3	35.2	41.5	42.0	40.4	38.5	30.5	24.4	21.6
	5%	WB	13.9	14.8	16.6	18.5	19.1	18.6	19.4	20.0	18.9	18.0	17.2	15.3
		MCDB	20.3	23.2	25.6	28.6	35.5	41.5	42.1	41.1	38.8	31.5	24.3	20.9
	10%	WB	12.7	13.6	15.5	17.7	18.4	18.1	18.6	19.3	18.3	17.1	16.3	14.2
		MCDB	19.8	22.7	25.1	28.8	35.7	41.1	41.7	41.2	38.6	32.4	24.1	20.5
Mean Daily Temperature Range	5% DB	MDBR	12.2	12.7	13.1	13.4	14.2	15.4	15.3	15.4	15.7	15.2	12.5	12.1
		MCDBR	15.2	15.1	15.6	15.4	15.5	16.2	15.7	15.9	16.1	16.2	15.3	14.7
		MCWBR	7.3	6.9	6.5	5.4	5.5	5.8	5.7	5.7	6.0	6.3	6.5	6.5
	5% WB	MCDBR	11.8	12.6	12.7	12.9	13.8	15.4	15.2	15.0	15.2	13.8	11.0	10.4
		MCWBR	6.4	6.4	6.0	4.9	5.0	5.7	5.9	5.9	6.0	6.1	4.9	5.2
Clear-Sky Solar Irradiance	taub		0.429	0.474	0.543	0.627	0.649	0.605	0.584	0.575	0.519	0.455	0.467	0.427
	taud		1.974	1.846	1.690	1.563	1.522	1.580	1.648	1.666	1.761	1.908	1.888	1.976
	Ebn at Noon		806	798	765	713	696	723	738	742	774	805	758	787
	Edh at Noon		161	196	241	280	291	273	254	249	221	181	174	155
All-Sky Solar Radiation	RadAvg		4.27	5.13	5.90	6.41	7.17	7.85	7.51	7.13	6.55	5.68	4.37	4.04
	RadStd		0.30	0.29	0.34	0.33	0.28	0.25	0.17	0.16	0.15	0.22	0.32	0.27

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (1 neighbor)	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	+103

Nomenclature: See separate page